Dimensionality Reductions that Preserve Volumes and Distance to Affine Spaces, and their Algorithmic Applications

Avner Magen *

Abstract

Let X be a subset of n points of the Euclidean space, and let $0 < \epsilon < 1$. A classical result of Johnson and Lindenstrauss [JL84] states that there is a projection of X onto a subspace of dimension $O(\log n/\epsilon^2)$, with distortion $\leq 1+\epsilon$. Here we show a natural extension of the above result, to a stronger preservation of the geometry of finite spaces. Basically, by a k-fold increment of the number of dimensions used compared to [JL84], a good preservation of volumes and of affine distances is achieved. Specifically, we show it is possible to embed a subset of size n of the Euclidean space into a $O(k\,(\max\{k, 1/\epsilon\})^2 \log n)$ dimensional Euclidean space, so that no set of size $s \leq k$ changes its volume by more than $(1+\epsilon)^{s-1}$. Moreover, distances of points from affine hulls of sets of at most $k-1$ points in the space are also not changed by more than a factor of $1+\epsilon$. A consequence of the above with $k=3$ is that angles can be preserved using asymptotically the same number of dimensions as is used in [JL84]. Our method can be applied to many problems with high-dimensional nature such as *Projective Clustering* and *Approximated Nearest Affine Neighbor Search.* In particular, it shows a first poly-logarithmic query time approximation algorithm to the latter.

1 Introduction

The dimension of a normed space that accommodates a finite set of points plays a critical role in the way this set is analyzed. The running time of most geometric algorithms is at least linear in the dimensionality of the space, and in many cases exponential in it. To represent the metric of n points in the Euclidean space, one clearly needs no more than $n-1$ dimensions. By relaxing the notion of isometry to near-isometry, the underlying structure of these points is well represented in a space with a much smaller number of dimensions. In their seminal paper [JL84], Johnson and Lindenstrauss show that a far more efficient representations capture almost precisely the metric nature of such sets. They present a very simple and elegant principle that allows one to embed such an n-point set into a t-dimensional Euclidean, with t merely $O(\epsilon^{-2}\log n)$, while preserving the pairwise distances to within a relative error of ϵ. One simply needs to project the original space onto a random t-dimensional subspace (and scale accordingly), and the above applies with high probability. Their argument therefore supplies a probabilistic algorithm for producing such an embedding.

Modifications that relate to the exact way the randomization is applied, and further improvements in the parameters were later proposed. For example it was shown that by using a projection onto t independent random unit vectors, a similar result is achieved. Lately, Achlioptas showed yet a simpler probability space for the desired embedding, by projecting the space onto random

*NEC Research Institute, Princeton, NJ. Part of this work was done while at the Hebrew University in Jerusalem. E-mail: avner@research.nj.nec.com.

vectors in $\{-1,1\}^N$ random vectors [Ach01]. In [EIO02], a derandomization of the probabilistic projection is given, leading to an efficient deterministic algorithm for finding such low-dimensional embeddings. Among the other simpler proofs to the result of Johnson and Lindenstrauss (to which we sometime refer as the *JL-lemma*) are [FM88, IM98, LLR95, DA99, AV99].

For metric spaces (X, d_X) and (Y, d_Y), and an embedding $f : X \to Y$ we define the distortion of f by $\sup_{x,y\in X} \frac{d_Y(f(x),f(y))}{d_X(x,y)} \cdot \sup_{x,y\in X} \frac{d_X(x,y)}{d_Y(f(x),f(y))}$. By this definition a good embedding is one that preserves the pairwise distances. However, a set X of points in the Euclidean space has many more characteristics in addition to the metric they represent. For example consider the center of gravity of a set of points and its average distance to them, the angles defined by triplet of points, volumes of sets, and distances between points to lines, planes and higher dimensional affine spaces that are spanned by subsets of X. Regarding (some of) these characteristics as part of the structure of X, we redefine the quality of an embedding to be one that preserves both the volumes of (certain) sets and the affine distance from them.

Recall that a volume of a set of points in an Euclidean space of size k is the $(k-1)$-dimensional volume (Lebesgue measure) of its convex-hull. For $k = 2$ this is just the distance between the points. For $k = 3$, this is the area of the triangle with vertices that are the three points of the set, etc. Throughout this paper we denote the volume of a set S in the Euclidean space by $\mathrm{Evol}(S)$. When considering a general metric space (not necessarily Euclidean), it is not a-priori clear whether there is a reasonable way to define a volume. Previous work, initiated by Feige [Fei00], defined this concept in such metric spaces, and measured the quality of an embedding from general metric spaces into Euclidean spaces. The volume preservation there applied to two different definition of volumes, the one in general metric spaces, and the one in Euclidean space. This line of work led to important algorithmic applications, most notably a polylogarithmic approximation algorithm for the bandwidth problem [Fei00], and an approximation algorithm to a certain VLSI layout problem [Vem98]. Our attention focuses on the case where both the original and the image space are Euclidean, and consequently the volume preservation concept is a well defined one and need not use the sophisticated definition of [Fei00]. Accordingly, our result should not be confused with results in the aforementioned framework, most notably Rao's result [Rao99], that deals with Euclidean metric, but with respect to its metric structure alone.

We say that a mapping that does not expand distances $f : \mathbb{R}^N \to \mathbb{R}^t$ distorts the volume of a set $S \subseteq \mathbb{R}^N$ of size k by $\left(\frac{\mathrm{Evol}(S)}{\mathrm{Evol}(f(S))}\right)^{\frac{1}{k-1}}$. The exponent in this expression should be thought of as a reasonable normalization measure (and was introduced in [Fei00]). We later mention the state of affairs when no such normalization is used.

Our Result: if X is an n point subset of $\mathbb{R}^N$, $0 < \epsilon < \frac{1}{3}$ and $t = 256k \log n \left(\max\{k, 1/\epsilon\}\right)^2$, there is a mapping of $\mathbb{R}^N$ into $\mathbb{R}^t$ that (i) does not distort the volume of sets of size at most k to by more than a factor of $1+\epsilon$. (ii) preserves the distance of points from affine hulls of sets of size at most $k-1$ to within a relative error of ϵ. We then remark that when $t \geq 256\epsilon^{-2}k^3 \log n$, there is a mapping of $\mathbb{R}^N$ into $\mathbb{R}^t$ under which (unnormalized) volumes of sets of size at most k do not change by more than a factor of $1+\epsilon$.

We argue that by using projections onto moderately larger dimensional subspaces (respectively, using larger number of independent unit vectors, etc.), we get the desired preservation of volumes and affine distances. For the analysis we make use of the following twist: by increasing the number of dimensions, compared to those needed in JL-lemma, we can preserve distances of a larger set than the original one, X. The crucial point is that we can carefully define a superset Z of X so that (i) the preservation of pairwise distances on Z enforces volume preservation and affine distance

preservation for X, and (ii) Z is not too big.

The setting of the Johnson Lindenstrauss embedding we use Of the many different formulations and proof-techniques to JL-lemma, we take the following one, which is a direct application of [FM88].

Theorem 1: *(paraphrasing Johnson and Lindenstrauss) Let $0 < \epsilon < \frac{1}{3}$, let X an n-point set in a Euclidean space, and let $t \geq 16 \log n/\epsilon^2$. Then there is an embedding of X into $\mathbb{R}^t$ with distortion $\leq 1+\epsilon$. The above embedding can be found in random polynomial time.*

Applications In their influential paper [IM98], Indyk and Motwani describe the way projections can be used for designing efficient algorithms for the *Approximate Nearest Neighbor* problem. The generalization of this problem from a set of points to a set of k-dimensional affine spaces is the *Approximate Nearest affine neighbor*. For $k = 1$, i.e., this is the problem of finding the closest line to a query point. This is a natural proximity problem, that has appeared few times in the literature. Combining Meiser's result for point location in arrangements of hyperplanes, with our result yields a poly-logarithmic query-time approximation to the problem [1], which is to the best of our knowledge the first. In 5.1 we show how exactly this can be achieved. Another classical problem in computational geometry that stems from data mining, might also use our result. Consider a data set in $\mathbb{R}^N$, where the "true dimensionality" is anticipated to be much smaller than N. In other words, it is assumed to be possible to cover (up to proximity) the set by a small number of k-dimensional affine spaces. The problem is known to be NP-hard even for dimension 2. Currently, there are approximation algorithms for the cases of two and three dimensions. For the higher dimensions, our method provides a way to reduce the problem to $k\left(\max\{k, 1/\epsilon\}\right)^2 \log n)$ dimensions.

There are also straight forward applications of our result to other problems which extend metrical questions to ones that consider volumes. In particular, the problem of finding the diameter of a set, can be extended to finding the biggest volume k-set. It is immediate by our result that this can be approximated in dimension which is logarithmic in the number of points.

2 Preliminaries and Notation

The norm $\|\cdot\|$ always stands for the Euclidean norm. We say that an embedding $\phi : X \to \mathbb{R}^N$ is a *contraction* if for every $x, y \in X$, $\|\phi(x) - \phi(y)\| \leq \|x - y\|$. Whenever the dimensionality of an Euclidean space is immaterial, we call it $\mathbb{R}^N$ without explicitly defining N. We will sometime refer to an affine space as a *flat*. An affine subspace of $\mathbb{R}^N$ that is spanned by points of $X \subseteq \mathbb{R}^N$ will be called X-flat (analogously, we define X-lines, X-planes, X-k-dimensional flats, etc.).

For a set $S \subseteq \mathbb{R}^N$ of size ν, we denote by $\mathcal{L}(S)$ the affine-hull of S, that is $\mathcal{L}(S) = \{\sum_{i=1}^{\nu} \lambda_i a_i | \sum_i \lambda_i = 1\}$. For a set $S \subseteq \mathbb{R}^N$ and $x \in \mathbb{R}^N$ we define $P(x, S)$ to be the projection of x onto $\mathcal{L}(S)$. The *affine distance* of x to S, $\mathrm{ad}(x, S)$, is defined to be the distance of x to the affine-hull of S, or equivalently $\|x - P(x, S)\|$. The affine distance will occasionally referred to as *height*. Let $r_1, r_2 \ldots, r_{\nu-2}$ be an arbitrary set of orthonormal vectors in $\mathcal{L}(S)$[2]. We now define the *corner* points of the pair (x, S).

[1] The approximation here is also in the sense that there is no guarantee for the answer to some small fraction of the query points

[2] We assume that S is an affinely independent set. Otherwise the volume of S or that of any set containing S is 0, and the volume of the image of such sets under a linear mapping will remain 0. For the affine distance estimates, we can just take an affinely independent set $S' \subseteq S$ for which $\mathcal{L}(S') = \mathcal{L}(S)$. We can therefore exclude sets which are affinely dependent from the discussion altogether.

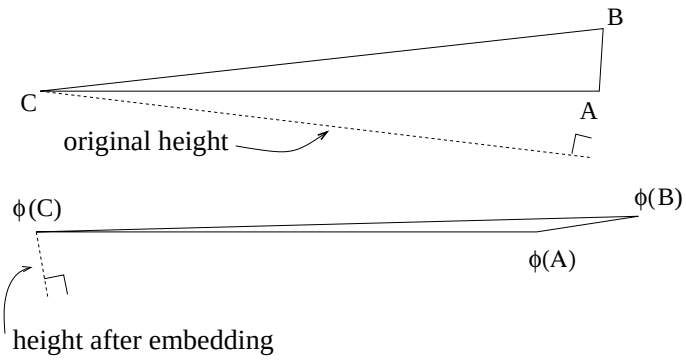

Figure 1: An example showing that areas and affine distances can dramatically change and practically vanish, even under a low-distortion embedding. Φ here can in fact be a projection.

The i-th corner point $\mathrm{cor}_i(x, S)$ is defined as $P(x, S) + \mathrm{ad}(x, S) \cdot r_i$, for $1 \leq i \leq \nu - 2$. An alternative way to define the corner points is saying they are the vertices of a (any) $\nu - 1$ dimensional cube, with an edge $x, P(x, S)$, and a facet supported by $\mathcal{L}(S)$. See figure 3.

3 Preserving Distances to Lines and Areas of Triangles

Consider the problem of finding a low-dimensional Euclidean embedding of a finite subset X of the Euclidean space, such that pairwise distances, areas from of triangles, and distance of points from X-lines do not change by much. Note that this case is a special case of the the general problem, as volumes of triplets of points are simply areas. This case is easier to analyze and moreover, the analysis of the general case uses some of the structure of the two dimensional case. The above case also gives a preservation result for angles as we later note.

The natural thing to try in order to reduce dimensionality and preserve geometrical features other than distances, is just to apply a random projection as in [JL84] to the point set X. It is desirable that such an embedding will not change such features as areas of triangles and distances of points to X-lines. Unfortunately, this naive approach will not work: consider a triangle with a very small angle. A low-distortion embedding can be applied to it, so that it is changed to a triangle with dramatically different angles, area and heights (see Figure 1). In fact there are *projections* with low distortion as in [JL84], that will lead to such a significant distortion of area and heights. The mapping Φ in Figure 1 can indeed be taken as such a projection. The following lemma shows that for a certain class of triangles, such a phenomenon is absent.

Lemma 2: *Let A, B, C be the vertices of an right angle isosceles triangle, where the right angle is at A, and let Φ be a contracting embedding of its vertices to a Euclidean space, such that the edges do not contract by more than $1 + \epsilon$, where $\epsilon < \frac{1}{3}$. Let H the length of $[AB]$, let $\vec{b}$ the vector $\Phi(B) - \Phi(A)$, and $\vec{c}$ be the vector $\Phi(C) - \Phi(A)$. Then*

1. $|\langle \vec{b}, \vec{c} \rangle| \leq 2\epsilon \cdot H^2$
2. $\frac{1-3\epsilon^2}{1+\epsilon} \cdot H \leq \mathrm{ad}(\Phi(C), \{\Phi(A), \Phi(B)\}) \leq H$.

Proof: Let $a = \|\vec{b} - \vec{c}\|$, and let, θ be the angle between $\vec{b}$ and $\vec{c}$. Now, $|\langle \vec{b}, \vec{c} \rangle| = |\|\vec{b}\|^2 + \|\vec{c}\|^2 - a^2|/2$. A simple analysis shows that this quantity is maximized (while satisfying the conditions on Φ) when

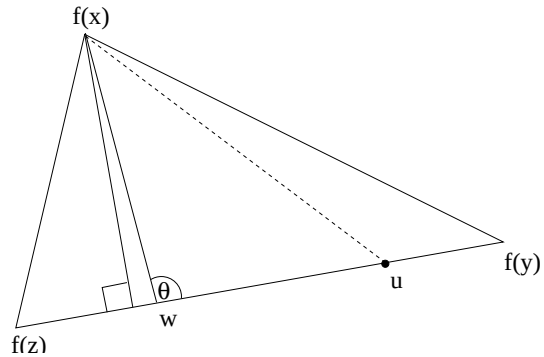

Figure 2: The triangle $f(x), w, u$ is the f-image of a right angle isosceles triangle

$\|\vec{b}\| = \|\vec{c}\| = \frac{H}{1+\epsilon}$ and $a = \sqrt{2}H$. Hence $|\langle \vec{b}, \vec{c} \rangle| \leq \frac{1}{2}H^2 \cdot (2 - 2/(1+\epsilon)^2) \leq 2\epsilon \cdot H^2$. It can also be verified that these values of $|a|, \|\vec{b}\|, \|\vec{c}\|$ also maximize $|\cos\theta| = |\|\vec{b}\|^2 + c^2 - a^2|/(2\|\vec{b}\|\|\vec{c}\|)$. Thus $|\cos\theta| \leq |(2 - 2(1+\epsilon)^2)/2| = 2\epsilon + \epsilon^2$, and accordingly $\sin\theta = \sqrt{1 - \cos^2\theta} \geq \sqrt{1 - 4\epsilon^2 - 4\epsilon^3 - \epsilon^4} > 1 - 3\epsilon^2$, for $\epsilon < \frac{1}{3}$. Finally,

$$H \geq \mathrm{ad}(\Phi(C), \{\Phi(A), \Phi(B)\}) = c \sin\theta \geq \frac{1 - 3\epsilon^2}{1+\epsilon} \cdot H.$$

■

In order to conclude that under a low-distortion embedding of the set X, areas and affine distances do not change by much, one would like to eliminate bad cases such as those in Figure 1. Think of the following physical model: Edges are rubber rods that can slightly contract due to a shock. Figure 1, shows that this shock may very well change the areas and heights of the triangle significantly. The remedy we propose is to supplement this rubber triangle with some additional rods to keep it stable. If the contraction of these rods is also limited, then placing them in appropriate locations, will eliminate cases such as the mapping Φ in Figure 1. Adding points on the edges of a triangle, has the effect of these rods when the distortion of the pairwise distances of the (new) set of points is limited.

Our strategy is therefore choosing points such that nice triangles as in Lemma 2 will emerge. This, together with the fact the embeddings we consider are linear, enables us to bound the changes of the heights. Area preservation is then immediately implied. To this end, we add to X auxiliary points, obtaining Y, so that every interval from a point in X to its projection on an X-line, is an edge of a straight angle isosceles triangle in Y. Finally, a linear low-distortion embedding of $\mathbf{Y}$ will be one that will not changes areas and affine distances by much for the original space X.

Theorem 3: *Let $\epsilon < \frac{1}{3}$ and let n, t be integers for which $t \geq 48\epsilon^{-2}\log n$. Then for any n-point subset X of the Euclidean space $\mathbb{R}^N$, there is a mapping $f : \mathbb{R}^N \to \mathbb{R}^t$, under which the areas of triangles in X are preserved to within a factor of $\frac{(1+\epsilon)^2}{1-3\epsilon^2}$, the distances of points from lines spanned by pairs of points of X are not changed by more than a factor of $\frac{1+\epsilon}{1-3\epsilon^2}$, and the sine of angles of triplet of points from X is preserved to within a factor of $\frac{(1+\epsilon)^2}{1-3\epsilon^2}$.*

Proof: For every set S of two elements of X, and every element $x \in X - S$, we consider two additional points: $P(x, S)$ and $cor_1(x, S)$, which is a point on $\mathcal{L}(S)$ whose distance from $P(x, S)$ is $\mathrm{ad}(x, S)$.

Let Y be the space that contains X and all the $2 \cdot 3\binom{n}{3}$ additional points, namely

$$Y = \bigcup_{|S|=2, S \subseteq X, x \in X-S} \{P(x,S), cor_1(x,S)\} \quad \bigcup X.$$

Since $|Y| \leq 6\binom{n}{3} + n \leq n^3$, and $t \geq 48\epsilon^{-2}\log n$, we have that $t \geq 16\epsilon^{-2}\log|Y|$. At this point Theorem 1 applies to the set Y, and so there is a contracting linear mapping f from Y to $\mathbb{R}^t$ with distortion $\leq 1+\epsilon$. We assume wlog that f is a contraction. We claim that f preserve affine distances of points in X to X-lines, and that it preserves areas of triangle with vertices in X.

Consider three different points $x, y, z \in X$. Let $w = f(P(x, \{y,z\})$ and $u = f(cor_1(x, \{y,z\}))$. Now let θ be the angle between $f(x) - w$ and $f(y) - w$ (see Figure 2). First observe that by linearity of f, the collinearity of $z, P(x, \{y,z\}), cor_1(x, \{y,z\}), y$ implies the collinearity of their f-images, namely $f(z), w, u, f(y)$. This means that $\mathrm{ad}(f(x), \{w,u\}) = \mathrm{ad}(f(x), \{f(z), f(y)\})$. We now apply the second assertion of Lemma 2 with $P(x, \{y,z\}), cor_1(x, \{y,z\})$ and x for A, B and C (note that these three are points in Y) to get that

$$\mathrm{ad}(x, \{y,z\}) \cdot \frac{1-3\epsilon^2}{1+\epsilon} \leq \mathrm{ad}(f(x), \{f(z), f(y)\}) \leq \mathrm{ad}(x, \{y,z\}).$$

Now,

$$\frac{1-3\epsilon^2}{(1+\epsilon)^2} \leq \frac{\|f(y)-f(z)\|}{\|y-z\|} \cdot \frac{\mathrm{ad}(f(x), \{f(y), f(z)\})}{\mathrm{ad}(x, \{y,z\})} \leq 1,$$

and by observing that

$$\frac{\mathrm{Evol}(f(S))}{\mathrm{Evol}(S)} = \frac{\|f(y)-f(z)\|}{\|y-z\|} \cdot \frac{\mathrm{ad}(f(x), \{f(y), f(z)\})}{\mathrm{ad}(x, \{y,z\})},$$

the result for the area preservation is also achieved. Finally, let α be the angle between $[yx]$ and $[yz]$, and let α' be the angle between $[f(y)f(x)]$ and $[f(y)f(z)]$. We have that

$$\frac{1-3\epsilon^2}{1+\epsilon} \leq \frac{\sin\alpha'}{\sin\alpha} = \frac{\|y-x\|}{\|f(y)-f(x)\|} \cdot \frac{\mathrm{ad}(f(x), \{f(y), f(z)\})}{\mathrm{ad}(x, \{y,z\})} \leq 1+\epsilon.$$

■

4 The General Case

Analogously to the $k = 3$ case, we add points to the original space, in such a way that the distance preservation of the extended space will imply an affine distance preservation and volume preservation in the original space. In the sequel, we restrict our attention to the preservation of affine distances, since the volume preservations will be a pretty simple consequence of that. The key idea in the extension of the previous construction will be substituting the right angle isosceles triangle by a simplex spanned by orthogonal vectors of equal size. The analysis, however, will be considerably more involved. In particular, for the analogue of Lemma 2 a simple maximization and an analysis of an angle-distortion cannot be made. Our analysis revolves around the simple relation $\mathrm{Evol}(V_{s-1}) = H \cdot \mathrm{Evol}(V_{s-2})/(s-1)$, where V_{s-1} is a simplex on s points, V_{s-2} the facet of V_{s-1} opposite one of the points, and H is the distance from that point to the facet. By the linearity of our embeddings, and having added appropriate auxiliary points, it is then possible to 'assume' for the analysis of H that V_{s-1} is the image of the nice simplex described above (refer

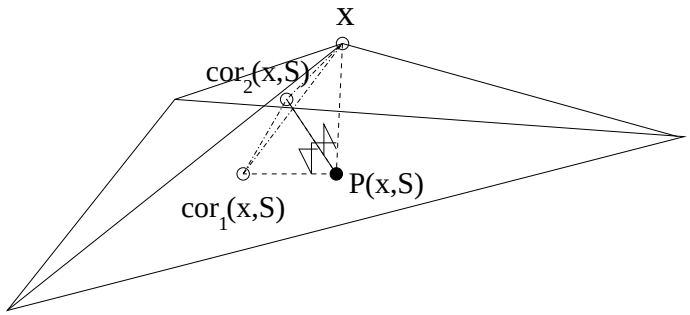

Figure 3: The height $[x, P(x,S)]$ is common to the original simplex (solid lines) and to the 'nice' simplex (dashed lines).

again to Figure 3 for the exposition). The actual estimate $\frac{\mathrm{Evol}(V_{s-1})}{\mathrm{Evol}(V_{s-2})}$ is then achieved by means of analysis of determinants of matrices with constraints resulting from the low distortion embedding on the vertices of V_{s-1} (Lemma 4). Eventually, the volume bounds will be achieved by an iterative use of the relation $\mathrm{Evol}(V_{s-1}) = H \cdot \mathrm{Evol}(V_{s-2})/(s-1)$.

For a matrix M, we let $\|M\|_\infty$ denote the maximal absolute value of the elements of M.

Lemma 4: *Let $\mu \leq \frac{1}{2(s-1)}$, and let A be a real $s \times s$ matrix, such that $\|A - I\|_\infty \leq \mu$. Denote by B the principal minor of A that is obtained by removing its last row and column. Then $1 - 2\mu \leq \det(A)/\det(B) \leq 1$.*

Proof: We first observe that if $y \in \mathbb{R}^s$ and $v = By$, then $\|v\|_\infty \geq \frac{1}{2}\|y\|_\infty$: Assume wlog that $|y_1| = \|y\|_\infty$. Now,

$$|v\|_\infty \geq |v_1| = |\sum_{i<s} y_i a_{1,i}| = |y_1 + \sum_{i<s} y_i(a_{1,i} - \delta_{1,i})|$$

$$\geq |y_1| - \mu \sum_{i<s} |y_i| \geq |y_1| - \mu(s-1)|y_1| \geq \frac{1}{2}|y_1| = \frac{1}{2}\|y\|_\infty$$

($\delta_{i,j}$ denotes the Kronecker Delta, which is 1 if $i = j$ and 0 otherwise). One immediate consequence of the above is that B is nonsingular, since it implies that if v is the zero vector then so is y.

Now, let v be the vector $(a_{1,s}, a_{2,s}, \ldots, a_{s-1,s})$. For every $j < s$, let $B^{(j)}$ be the matrix B with the j-th column replaced by v. We next argue that $|\frac{\det(B^{(j)})}{\det(B)}| \leq 2\mu$ for all $1 \leq j < s$. Let $y_j = \frac{\det(B^{(j)})}{\det(B)}$. Recall that by Cramer's rule, $y = (y_1, y_2, \ldots, y_{s-1})$ is the (unique) solution to $By = v$. We now get $|\frac{\det(B^{(j)})}{\det(B)}| = |y_j| \leq \|y\|_\infty \leq 2\|v\|_\infty = 2\max_{i<s} a_{i,s} \leq 2\mu$.

Now,

$$\det(A) = a_{s,s}\det(B) + \sum_{i<s}(-1)^{i+s} a_{s,i}\det(B^{(i)}).$$

Therefore

$$\frac{\det(A)}{\det(B)} = a_{s,s} + \sum_{i<s}(-1)^{i+s} a_{s,i}\frac{\det(B^{(i)})}{\det(B)} \geq 1 - \mu - \mu(s-1)\cdot 2\mu \geq 1 - 2\mu.$$

■

We now use the above algebraic fact to deduce the desirable geometric bounds. The flavor of the analysis to follow is this: Consider a set of points $p_0, p_1, \ldots, p_s$ in $\mathbb{R}^N$, and let $v_i = p_i - p_0$ for

$i > 0$. A standard fact is that the volume of the convex hull of $p_0, p_1, \ldots, p_s$ is the square root of the determinant of the matrix of inner products of the v_is, divided by $(s-1)!$. The volume of the facet which is the convex hull of $p_0, , p_1, \ldots, p_{s-1}$ is calculated in exactly the same way. The relevant matrix this time is just the principal minor of the first matrix mentioned. But under the right normalization, the ratio between these two volumes is just the affine distance of p_s to $\mathcal{L}(p_0, , p_1, \ldots, p_{s-1})$.

Definition 5: We call a set of vectors $v_1, v_2, \ldots v_\nu$ a (c, μ) *almost orthonormal* system if for all $i \neq j$, $-\mu \leq \frac{\langle v_i, v_j \rangle}{c^2} \leq \mu$, and $1 - \mu \leq \frac{\|v_i\|^2}{c^2} \leq 1 + \mu$. Note that a $(1, 0)$ *almost orthonormal* system is simply a set of orthonormal vectors.

Claim 6: *Let $p_0, p_1, \ldots, p_s$ be points in $\mathbb{R}^N$ such the $\{p_i - p_0\}_{i \geq 1}$ is a (c, μ)* almost orthonormal *system of vectors where $\mu \leq \frac{1}{2(s-1)}$, and let H be $\mathrm{ad}(p_s, \{p_0, p_1, \ldots, p_{s-1}\})$. Then*

$$\sqrt{1 - 2\mu} \cdot c \leq H \leq c.$$

Proof: Assume wlog that $c = 1$. Otherwise scale the space down by a factor of c, and the desired result applies from this reduced case. Let Q_r be the convex hull of $p_0, p_1, \ldots, p_r$, let S_r be the $r \times s$ matrix whose rows are $p_1 - p_0, p_2 - p_0, \ldots, p_r - p_0$, and let T_r be the $r \times r$ matrix which is the Graham product of S_r, that is $T_r = S_r S_r^t$. Using the two previously mentioned relations $\mathrm{Evol}(Q_r) = \sqrt{\det(T_r)}/(r-1)!$, and $H = (s-1) \cdot \mathrm{Evol}(Q_s)/\mathrm{Evol}(Q_{s-1})$, we get

$$H = \sqrt{\det(T_s)/\det(T_{s-1})}.$$

The fact that $\{p_i - p_0\}_i \geq 0$ is a $(1, \mu)$ *almost orthonormal* system is equivalent to $\|T_r - I\|_\infty \leq \mu$. Hence the conditions of Lemma 4 hold with $A = T_s$ and $B = T_{s-1}$. We can now conclude that

$$\sqrt{1 - 2\mu} \leq H = \sqrt{\det(T_s)/\det(T_{s-1})} \leq 1.$$

■

Remark 7: It is interesting to note that the condition $\mu \leq \frac{1}{2(s-1)}$ in both Lemma 4 and in Claim 6 cannot be relaxed by much: let $p_i = e_i$ for $1 \leq i \leq s$, where e_i is the i'th standard vector in $\mathbb{R}^N$, and let $p_0 = \frac{1}{s} \sum_{i \geq 1} e_i$. Then it is easy to check that $\{p_i - p_0\}_{i \geq 1}$ is a $(1, \frac{1}{s-1})$ *almost orthonormal* system of vectors. But here the p_i-s are affinely dependent, implying $H = 0$!

4.1 The Main Theorem

Theorem 8: *Let $\epsilon < \frac{1}{3}$ and let k, n, t be integers greater than 1, for which $t \geq 256k \log n \left(\max\{k, 1/\epsilon\}\right)^2$. Then for any n-point subset X of the Euclidean space $\mathbb{R}^n$, there is a linear mapping $f : X \to \mathbb{R}^k$, such that for all subsets T of X, $1 < |T| \leq k$,*

$$\mathrm{Evol}(T)/(1+\epsilon) \leq \mathrm{Evol}(f(T)) \leq \mathrm{Evol}(T),$$

and for $x \in T$,

$$\mathrm{ad}(x, T - \{x\})/(1+\epsilon) \leq \mathrm{ad}(f(x), f(T - \{x\})) \leq \mathrm{ad}(x, T - \{x\}).$$

Proof: Let

$$Z = \bigcup_{S \subseteq X, 1<|S|<k, x \in X-S} \{P(x,S), cor_1(x,S), cor_2(x,S), cor_{|S|-1}(x,S)\} \bigcup X,$$

and set $\Delta = \frac{1}{4}\min\{\frac{1}{k}, \epsilon\}$. We claim that there is a contracting linear mapping $f : Z \to \mathbb{R}^t$ with distortion $\leq 1+\Delta$. To see that observe that since

$$|Z| \leq n + \sum_{i=3}^{k} (i-1)^2 \binom{n}{i} \leq n^k,$$

we get that

$$t \geq 256k \log n \left(\max\{k, 1/\epsilon\}\right)^2 \geq 16 \log |Z|/\Delta^2,$$

whence JL-lemma applies.

Next, we show that all distances of points to s-dimensional X-flats are preserved to within a relative error of ϵ. Let S be a set of size s, where $1 < s < k$, and $x \in X - S$, and let $p_0 = f(P(x,S)), p_i = f(cor_i(x,S))$ for $0 < i < s$, and $p_s = f(x)$ (observe that $p_i \in f(Y)$). Now, for $i < j$, we apply Lemma 2, with $\vec{b} = p_i - p_0, \vec{c} = p_j - p_0$, and since f is a contracting embedding with distortion $\leq 1+\Delta$, we get $|\langle p_i - p_0, p_j - p_0\rangle| \leq 2\Delta \cdot \mathrm{ad}(x,S)^2$. For $i = j$, we use again this property of f: $\frac{\mathrm{ad}(x,S)}{1+\Delta} \leq \|p_i - p_0\| \leq \mathrm{ad}(x,S)$, to get $(1-2\Delta)\cdot \mathrm{ad}(x,S)^2 \leq \left(\frac{1}{1+\Delta}\right)^2 \cdot \mathrm{ad}(x,S)^2 \leq \|p_i - p_0\|^2 \leq \mathrm{ad}(x,S)^2$. Therefore $\{p_i - p_0\}_{i \geq 1}$ is a $(\mathrm{ad}(x,S), 2\Delta)$ *almost orthonormal* system of vectors, and since $2\Delta = \frac{1}{2}\min\{\frac{1}{k}, \epsilon\} \leq \frac{1}{2(k-1)} \leq \frac{1}{2(s-1)}$, we can apply Claim 6:

$$\sqrt{1-4\Delta} \cdot \mathrm{ad}(x,S) \leq \mathrm{ad}(f(x), \{p_0, p_1, \ldots, p_{s-1}\}) \leq \mathrm{ad}(x,S).$$

Since $\mathcal{L}(S) = \mathcal{L}(\{P(x,S), cor_1(x,S), \ldots, cor_{s-1}(x,S)\})$, and since f is linear, we get

$$\mathrm{ad}(f(x), \{p_0, p_1, \ldots, p_{s-1}\}) = \mathrm{ad}(f(x), f(S)).$$

Relating Δ and ϵ we get $\sqrt{1-4\Delta} \geq \sqrt{1-\epsilon} \geq \frac{1}{1+\epsilon}$, from which the first part of the theorem is proved, namely

$$\frac{1}{1+\epsilon} \cdot \mathrm{ad}(x,S) \leq \mathrm{ad}(f(x), f(S)) \leq \mathrm{ad}(x,S).$$

We now turn to the volume of the set in the small-dimensional space. Using the bound in the affine distances, this task becomes quite straight forward. Let $S_r = x_1, x_2, \ldots, x_r$. It is known that

$$\mathrm{Evol}(S_r) = \frac{1}{(r-1)!} \cdot \prod_{i=1}^{r-1} \mathrm{ad}(x_{i+1}, S_i),$$

and so the volume distortion of S, $(\mathrm{Evol}(S)/\mathrm{Evol}(f(S)))^{\frac{1}{r-1}}$ is just the geometric mean of

$$\{\mathrm{ad}(x_{i+1}, S_i)/\mathrm{ad}(f(x_{i+1}), f(S_i))\}_{i=1}^{r-1},$$

and we can conclude

$$1 \leq (\mathrm{Evol}(S)/\mathrm{Evol}(f(S)))^{\frac{1}{r-1}} \leq 1+\epsilon.$$

Substituting ϵ in the proof of Theorem 8 by $\frac{4}{5}\epsilon/k$ yields

$$1 \leq \mathrm{Evol}(S)/\mathrm{Evol}(f(S)) \leq (1 + \frac{4}{5}\epsilon/k)^{k-1} \leq e^{.8\epsilon} \leq 1+\epsilon$$

for $\epsilon < \frac{1}{3}$. In this setting, $\Delta = \frac{1}{4}\min\{\frac{1}{k}, \frac{4}{5}\epsilon/k\} = \frac{1}{5}\epsilon/k$. Therefore $t = O(k\Delta^{-2}\log n) = O(\epsilon^{-2}k^3 \log n)$ dimensions suffice, if the volumes themselves (i.e. not only their normalizations) are to be preserved to within a relative error of ϵ. ■

5 Applications

5.1 The Approximated Nearest Neighbor to Affine Spaces problem

Let $\mathcal{F}_1, \mathcal{F}_2, \ldots, \mathcal{F}_n$ be n k-dimensional flats in $\mathbb{R}^d$, and let $x \in \mathbb{R}^d$ be a query point. To answer a Nearest Neighbor to Affine Spaces is to find the flat closest (in the Euclidean sense) to x. In [Mei93] Meiser presents a solution to the point location problem in arrangements of n hyperplanes in $\mathbb{R}^d$ with running time $O(d^5 \log n)$ and space $O(n^{d+1})$. That paper was a breakthrough in that it was the first time (and to the best of our knowledge also the last) where an algorithm to the problem was presented, that is not exponential neither in d nor in $\log n$. We show that by combining this result with ours, a considerable improvement to the problem of approximating the Nearest Affine Neighbor can be achieved. Let $\rho_i : \mathbb{R}^d \to \mathbb{R}$ be the functions that are the squares of distances from the flats, i.e. $\rho_i(x) = dist(x, \mathcal{F}_i)$. Note that ρ_i^2 are polynomials of degree 2 in $x_1, \ldots, x_d$. We now use the following standard linearization of such polynomials. We use the transformation $\xi : \mathbb{R}^d \to \mathbb{R}^{d'}$ which is the assignment transformation to all the monomials of degree at most two in $x_1, \ldots, x_d$. For example, if $d = 2$, $\xi(x) = \xi((x_1, x_2)) = (1, x_1, x_2, x1x_2)$. Clearly, $d' = \binom{d}{2} + 2d + 1$. Next, we introduce the functions ρ_i' defined by $\rho_i'(\xi(x)) = \rho_i^2(x)$. It is easy to see now that the ρ_i' are linear functionals [3]. We can now reduce the *Nearest Affine Neighbor* problem with query point x to the Vertical Ray Shooting problem from x: 'Shoot' a ray from $(\xi(x), -\infty)$ 'upwards', in other words to the positive direction of the last coordinate. Let $\mathcal{S}_i$ be (one of) the first surface this ray hits. Then $\mathcal{F}_i$ is the closest of $\mathcal{F}_1, \mathcal{F}_2, \ldots, \mathcal{F}_s$ to x. This vertical ray-shooting query in linear arrangements can be easily answered using the data-structure of Meiser.

The Preprocessing time, as well as space needed for the above are $O(n^{d'+1}) = O(n^{d^2})$, and the query time is $O(d'^5 \log n) = O(d^{10} \log n)$. We now apply our result in the following way. Think of each flat as a subset of $k+1$ points. We can embed these $O(nk)$ points to $O(k\,(\max\{k, 1/\epsilon\})^2 \log n)$ dimensions, and for k constant this is just $O(\log n/\epsilon^2)$. Concatenation of the above yields a query time of $O(\epsilon^{-20}(\log n)^1 1)$, and a preprocessing time (and space) of $n^O(\epsilon^{-4} \log^2 n)$. The standard problem in this approach remains: we need to answer queries on any point of the space, and not on a predetermined set. We here use the probabilistic formulation of the Johnson-Lindenstrauss embeddings, and omitting the technical details, we claim that by a further constant factor to the number of dimension, the embedding is good to all but exponentially small fraction of the points in the space (suppose we look at the Haar measure on the sphere for this quantification). We can therefore approximate the answer to all but a small fraction of the query points.

5.2 Projective Clustering

Projective Clustering is a well known problem which has important applications in data-mining. This special variant of clustering, relates very closely to the geometric structures that are discussed in this paper. Here is the problem: Given an n-point set X in $\mathbb{R}^N$ and an integer $s > 0$, find s k-dimensional flats (k-flat), $h_1, \ldots, h_k$, so that the greatest distance of a any point from its closest flat is minimized. In other words, this is a s-clustering problem, where a cluster is defined by a k-flat, and the quality of the clustering, its *width*, is the maximal distance of any point from its corresponding flat [4]. The "regular" clustering is in fact a special case, where $k = 0$. There are different ways to define the solution for the *projective clustering* problem: the optimal width, the clustering, and the k-flats themselves. Many works revolves around this problem, and as far as the general case is concerned, little is known about the ways to approximate the problem. In a recent

[3] The ρ_i' are no longer defined on the whole space, but this should not be of any concern to us.

[4] Any other variant, such as average distance, sum and sum-of-squares is probably as applicable here.

paper [PC00], Agarwal and Procopiuc, give an efficient approximation algorithm for the planar case where the flats are lines, i.e. $N = 2$ and $k = 1$. We show that under certain restrictions one can reduce the problem to one with a dimensionality of the space depending logarithmically in n, such that only a small inaccuracy incurs.

In [PC00], the original problem is reduced to the variant where the candidate flats are only the X-k-flats. Namely only affine-subspaces that are spanned by $k+1$ points from X are considered. We call this variant the *Median Projective Clustering.* Agarwal et al. show that by this reduction an additional factor of at most 2 is added to the approximation. We claim

Theorem 9: *An instace of the* Median Projective Clustering *with $X \subseteq \mathbb{R}^N$ and k the dimensionality of the flats, can be reduced to a t-dimensional space instance with a $1+\epsilon$ approximation to the optimal width, where $t = O(k\,(\max\{k, 1/\epsilon\})^2 \log n)$,.*

Proof: Using Theorem 8, we can map X to $\mathbb{R}^t$, such that the distance of points in X to k-dimensional X-flats do not expand, and do not contract by more than a factor of $1+\epsilon$ Therefore any solution w^* in the reduced problem corresponds to a solution w in the original problem, with $w^* \leq w \leq 2w^*$, and we immediately get that $\tau^* \leq \tau \leq 2\tau^*$, where τ, τ^* are the optimal solution to the original problem, and the optimal solution to the reduced problem respectively. Note that the clustering for that width can also be given from the reduced problem. It is not clear, however, how to reconstruct the k flats from the ones in the reduced problem, as the k-flats in the smaller space do not map back to the original space (a projection is not injective). ■

Acknowledgments I would to thank Robert Krauthgamer and Nati Linial for the discussions that led the question of this paper. To Piotr Indyk and Sariel Har Peled for the Computational-Geometry relevant Background and references, and many suggestions for Applications. In particular to Sariel, who pointed out the way to use Meiser's result in Section 5.1. Thanks are also due to Funda Ergun and Ronitt Rubinfeld for many insightful comments on an earlier manuscript.

References

[Ach01] Dimitris Achlioptas. Database-friendly random projections. In *roceedings of PODS 01*, pages 274–281, 2001.

[AV99] R.I. Arriaga and S. Vempala. An algorithmic theory of learning: Robust concepts and random projections. In *Proceeding of the 31th Symposium on the Theory of Computing*, New York, 1999.

[DA99] S. Dasgupta and Gupta A. An elementary proof of the johnson-lindenstrauss lemma. In *Technical Report TR-99-06*, Computer Scinece Institute, Berkeley, CA, 1999.

[EIO02] L. Engebretsen, P. Indyk, and R. O'Donnell. Derandomized dimensionality reduction with applications. In *Proceedings of the 13th Symposium on Discrete Algorithms.* IEEE, 2002.

[Fei00] Uriel Feige. Approximating the bandwidth via volume respecting embeddings. *J. Comput. System Sci.*, 60(3):510–539, 2000. 30th Annual ACM Symposium on Theory of Computing (Dallas, TX, 1998).

[FM88] P. Frankl and H. Mahera. The johnson lindenstrauss lemma and the sphericity of some graphs. *J. Combin. Theory Ser. B*, 44(2):355–362, 1988.

[IM98] P. Indyk and R. Motwani. Approximate nearest neighbor: towards removing the curse of dimensionality. In *Proceedings of the Thirty Second Symposium on Theory of Computing.* ACM, 1998.

[JL84] William B. Johnson and Joram Lindenstrauss. Extensions of Lipschitz mappings into a Hilbert space. In *Conference in modern analysis and probability (New Haven, Conn., 1982)*, pages 189–206. Amer. Math. Soc., Providence, RI, 1984.

[LLR95] N. Linial, E. London, and Yu. Rabinovich. The geometry of graphs and some of its algorithmic applications. *Combinatorica*, 15(2):215–245, 1995.

[Mei93] S. Meiser. Point location in arrangements of hyperplanes. *Information and Computation*, 106:286–303, 1993.

[PC00] Agarwal P. and Procopiuc C. Approximation algortihms for strip cover in the plane. In *Proceeding of the 11th ACM-SIAM Symposium on Discrete Algorithms*, pages 373–382, 2000.

[Rao99] Rao. Small distortion and volume preserving embeddings for planar and Euclidean metrics. In *Proceedings of the Fifteenth Annual Symposium on Computational Geometry (Miami Beach, FL, 1999)*, pages 300–306 (electronic), New York, 1999. ACM.

[Vem98] S. Vempala. Approximating vlsi layout problems. In *Proceedings of the thirty ninth Symposium on Foundations of Computer Science.* IEEE, 1998.